

*Teaching Notes 6B**

Beyond MM: Debt and Taxes

Pietro Veronesi
The University of Chicago
Booth School of Business

*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

Overview

- Review MM assumptions
 - Better understand effects of capital structure on a firm's total value
- Focus on first two assumptions
 - Perfect Capital Markets
 - No Taxes

Objectives

- Understand the “qualitative” aspects that are introduced when we relax MM assumptions
- Understand “quantitatively” what relaxing a particular assumption implies for capital structure

Recall: Modigliani-Miller (MM) Proposition

- *MM Proposition:*
 - *“The total value of the securities issued by a firm is independent of the firm’s choice of capital structure. The firm’s value is determined by its real assets and growth opportunities, not by the types of securities it issues.”*
- MM implies that capital structure does not matter for the value of the firm
- But MM holds only under certain assumptions

MM Assumptions (review)

- MM requires the following assumptions:
 - Perfect and complete capital markets
 - No Taxes
 - Bankruptcy is not costly
 - Capital Structure does not affect investment policy and cash-flows
 - Symmetric information

Assumption #1

Perfect and Complete Capital Markets

First Assumption: Perfect Capital Markets

- If capital markets are imperfect, the arbitrage argument does not work
 - Remember the Portfolio/No arbitrage proof
- The argument may fail on two fronts
 - *Existence of arbitrage (imperfection)*: Market may not price securities so that Law of One Price holds.
 - *Existence of clienteles (incompleteness)*: Investors cannot trade “as they wish” in the market.

Relaxing Perfect Capital Markets Assumption: Existence of Arbitrage (Imperfection)

- Market price may not reflect the risk of securities perfectly (for some reason)
 - If CEO believes value of the firm is determined totally by P/E or EPS, the firm could change capital structure to change P/E or EPS
- Why could P/E or EPS determine firm value?
 - Perhaps there may be some mispricing in the market...
 - ...but unlikely to be large or long-term
 - For us, projects financed over a long-term
 - Therefore, not likely to be first-order concern

Relaxing Perfect Capital Markets Assumption: Existence of Clienteles (Incompleteness)

- **What if capital markets limit investors' ability to trade?**
 - Investors cannot undo changes in capital structure by themselves
 - If a firm is slanting its capital structure to benefit from preferences of a set of investors → “catering to a Clientele”: *Clientele Effect*
- **Examples**
 - If investors cannot borrow → they can't “create own leverage” → investors who like risky returns will pay a premium for equity of more leveraged firms → firms boost value by leveraging up
 - Some investors may pay a premium for leverage
- **Typically not important for firms' capital structures.**
 - Investment banks (rather than firms) have a comparative advantage at completing the market
 - Investment banks make CLOs to fill demand for AAA debt; don't ask risky borrowers to become “less risky” by retiring debt!

Assumption #2

No Taxes

Relaxing Assumption of No Taxes

- MM assumes no taxes
 - With taxes, debt creates a debt tax shield
 - We know from APV and WACC that the tax shield is valuable
 - Thus, the presence of tax shields could make it desirable to have debt in the capital structure instead of equity
 - ➔ Value of the firm may be impacted by the choice of its financing
- Note that, in computing the debt tax shield, we have used the corporate tax rate τ_c
 - Let us take a closer look at why this is the case

Why Only Consider Corporate Tax?

- In practice, 5 tax rates to worry about
 1. Corporate ordinary rate (τ_c), paid by firms on their taxable earnings from operations
 2. Corporate capital gains rate ($\tau_{c, cg}$), paid by firms on their capital gains/losses
 3. Personal ordinary rate (τ_p), paid by individuals on income derived from wages, salaries, and interest
 4. Personal capital gains rate ($\tau_{p, cg}$), paid by individuals on profits they derive from selling equity that has appreciated
 5. Personal dividend tax rate ($\tau_{p, div}$), paid by individuals on dividends
- Usually we only worry about corporate taxes

Why Only Consider Corporate Tax?

Top Income Tax Rates Faced by Individuals and Corporations in the US

Time Period	C-Corporation		Individual		
	Ordinary Income	Capital Gains	Ordinary Income	Dividends	Capital Gains
	(τ_c)	($\tau_{c,cg}$)	(τ_p)	($\tau_{p,div}$)	($\tau_{p,cg}$)
Pre-1981	46.0%	28.0%	70.0%	70.0%	28.0%
1982-1986	46.0%	20.0%	50.0%	50.0%	20.0%
1987	40.0%	28.0%	39.0%	39.0%	28.0%
1988-1990	34.0%	34.0%	28.0%	28.0%	28.0%
1991-1992	34.0%	34.0%	31.0%	31.0%	28.0%
1993-1996	35.0%	35.0%	39.6%	39.6%	28.0%
1997-2000	35.0%	35.0%	39.6%	39.6%	20.0%
2001-2002	35.0%	35.0%	38.6%	38.6%	20.0%
2003-2012	35.0%	35.0%	35.0%	15.0%	15.0%
2013-2017	35.0%	35.0%	39.6%	20.0%	20.0%
2018-2019	21.0%	21.0%	37.0%	20.0%	20.0%

Sources: Scholes et al. (2015), Taxes and Business Strategy (5th ed), Table 1.1, p. 10

KPMG (2018), Tax Reform – KPMG Report on New Tax Law

Why Only Consider Corporate Tax?

	Flow to debt-holders	Flow to equity-holder
Assumed Earnings	\$1	\$1
Corporate taxes	0 (debt is deducted)	τ_c
Income after corporate taxes	1	$1 - \tau_c$
Personal taxes	τ_p	$\tau_{p,div}$
Income after taxes	$1 - \tau_p$	$(1 - \tau_{p,div})(1 - \tau_c)$

- $1 - \tau_p > (1 - \tau_{p,div})(1 - \tau_c)$ generally holds for parameters on previous slide, meaning that debt financing yields greater cash flow to investors than equity financing

Why Only Consider Corporate Tax?

$$PV(\text{Debt Tax Shield}) = D \left[1 - \frac{(1 - \tau_c)(1 - \tau_{p,div})}{(1 - \tau_p)} \right]$$

- Formula due to Miller (1977)
 - If $\tau_{p,div} < \tau_p$, the DTS becomes less valuable than indicated by our previous calculations
- Note that if $\tau_{p,div} = \tau_p$, this reduces to earlier formula, i.e.
 $PV[\text{DTS}] = \tau_c D$
 - Will often make this assumption

Why Only Consider Corporate Tax?

- Large investors tend to be institutional tax exempt investors
 - Pension funds, insurance companies, endowments
- Hence, usually ignore investors' personal taxes
 - ➔ Calculate DTS from corporate tax only

Optimal Capital Structure with Taxes

- If DTS is the only effect of capital structure, then firms should max out on leverage
 - However, negative EBIT reduces value, since it doesn't generate tax shield in that period
 - → want our interest payments to eliminate our EBIT (interest coverage = 100%)
 - **Based on average EBIT** → $D/(D+E)$ should be around **50-60%**
- **In reality**, firms have much less leverage:
 - → average $D/(D+E)$ is around **33%**
 - So there must be other effects as well
- One implication of this notion:
 - Growth firms should have lower D/E since their value is in future growth, not current EBIT

Capital Structure across Industries

Debt-to-Enterprise Value for Select Industries (Q2 2018)



Example

Example: Estimating Impact of Tax Shield

- In a levered recapitalization (a “recap”), a company borrows money and pays it to the shareholders
 - Either as a dividend or a share repurchase
- What is the effect of a “recap” on the share price? Who gains or loses?
 - Remember, we are now not in a MM world and allow for taxes

Example: Estimating Impact of Tax Shield

- Midco has 20m shares, trading at \$15 per share, and no debt
 - Midco has stable earnings
 - Tax-rate is 35%
 - Discount rate for tax-shield is r_D
- Now, Midco will borrow \$100m and use the proceeds to repurchase shares

Example: Estimating Impact of Tax Shield

- What is the effect on the share price?
- What is the value of the DTS?
- What is the value of Midco after recap?
- How many shares can we repurchase and at what price?
- Who gains and who loses from the recap?

Example: Estimating Impact of Tax Shield

- What is the value of the DTS?
 - $PV[DTS] = t \times D = \$35m$
- What is the value of Midco after recap?
 - Total assets = \$300m (old assets) + \$35m (tax shield)
- How many shares can we repurchase and at what price?
 - “Two-equations and Two-unknowns”
 - $P = \$16.75$ and $R = 5.97m$
- Who gains and who loses from the recap?
 - Old shareholders gain \$35m or \$1.75 per share
 - Debt holders don't care

Example: Estimating Impact of Tax Shield

- In a levered recap, the value of the equity increases by the value of the DTS
 - The shareholders receive the total increase in value generated by the DTS
 - Regardless of whether they sell their shares or not

Takeaways: MM – First Two Assumptions

- If markets are imperfect, there can be a clientele effect on capital structure
 - Less important for capital structure
- Personal taxes reduce the value of the DTS
 - But large institutional investors are not taxed differentially on income from equity and debt, so this effect is usually small
- If taxes were the only consideration for capital structure, firms would increase leverage as much as possible
 - In practice, don't see this → must be other effects we are missing